

NOTES
FOR
FINALS

AMPLITUDE SHIFT KEYING (ASK)

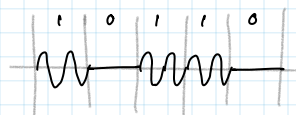
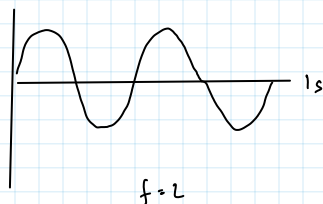
scheme

1: no change
0: 0

Terminology

carrier signal — OG signal

modulated signal — carrier + data

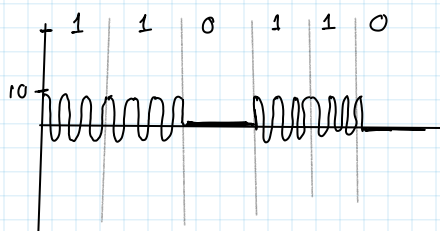


Carrier

A : 10

f : 3

p : 90



$$2^2 = 4$$

00 → 5V

01 → 10V

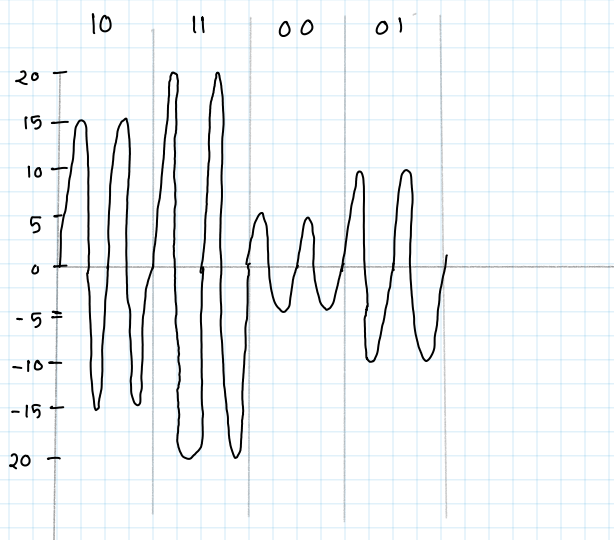
10 → 15V

11 → 20V

1011001

f = 2

p = 0°



$$f \rightarrow 3$$

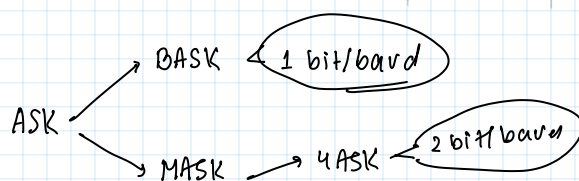
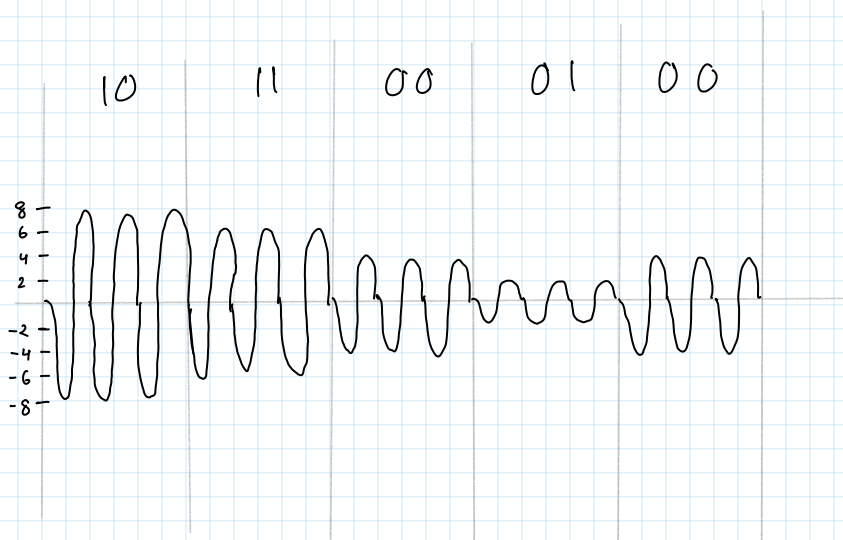
$$P \rightarrow 180^\circ$$

$$00 \rightarrow 4V$$

$$01 \rightarrow 2V$$

$$10 \rightarrow 8V$$

$$11 \rightarrow 6V$$



FREQUENCY SHIFT KEYING (FSK)

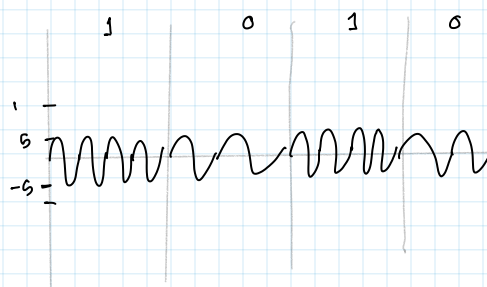
$$A : 5V$$

$$F : f_c = 5$$

$$P : 0^\circ$$

$$1 : f_1 = f_c + \Delta f = 4$$

$$0 : f_0 = f_c - \Delta f = 2$$



$$A = 10$$

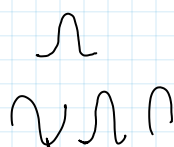
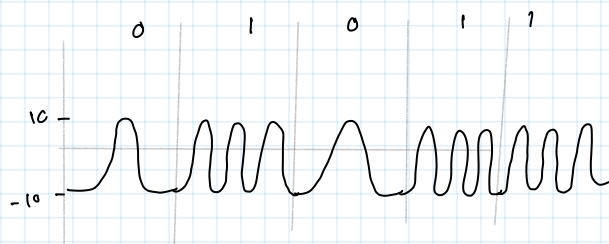
$$F = f_c = 2$$

$$P = 270^\circ$$

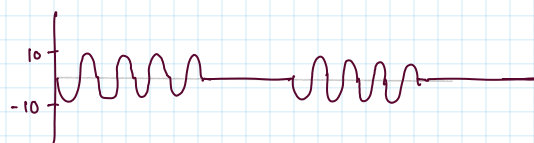
$$\Delta f = \frac{f_c}{2} = 1$$

$$1 : f_1 = 3$$

$$0 : f_0 = 1$$



1 0 1 0



$$\Delta f = \frac{f_c}{2} \rightarrow \text{Draw FSK}$$

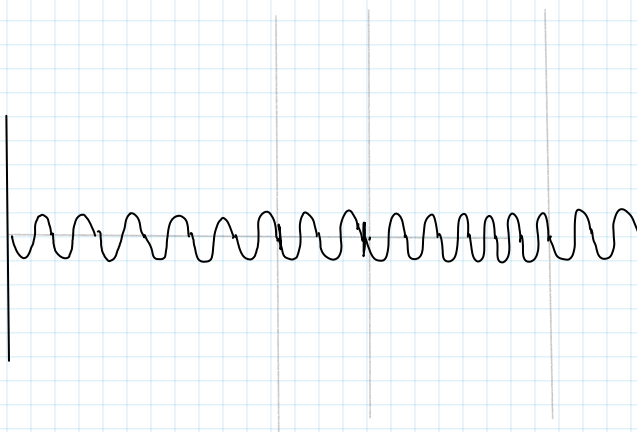
$$\Delta f = 2$$

$$f_c = 4$$

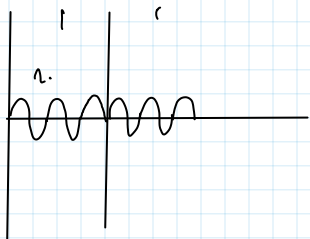
$$1: f_1 = 6$$

$$0: f_0 = 2$$

u



when decimal

1: 2.5
0: 1.5

4FSK

$$f_c = 10$$

$$\Delta f = 2$$

$$\begin{array}{l} 00 = 8 - 2 = 6 \\ 01 = 10 - 2 = 8 \\ \hline 10 = 10 + 2 = 12 \\ 11 = 12 + 2 = 14 \end{array} \rightarrow \begin{array}{l} \text{imaginary} \\ \text{divide (ur choice !)} \end{array}$$

$$f_c = 8$$

$$\Delta f = 3$$

$$00 \Rightarrow 5 - 3 = 2$$

$$01 \Rightarrow 8 - 3 = 5$$

$$10 \Rightarrow 8 + 3 = 11$$

$$11 \Rightarrow 11 + 3 = 14$$

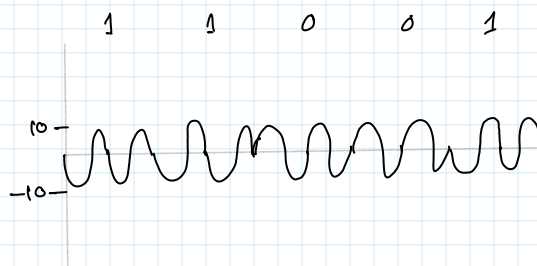
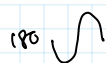
PSK

$A = 10$

$f = 2$

$1 \rightarrow \phi : 0^\circ$

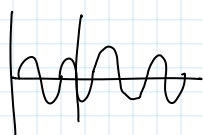
$0 \rightarrow \phi = 180^\circ$

 180° 

Note: always diff in phase betⁿ bits have to be 180°

↳ oscilloscope just inverts.

coherent
FSK



non-coherent FSK



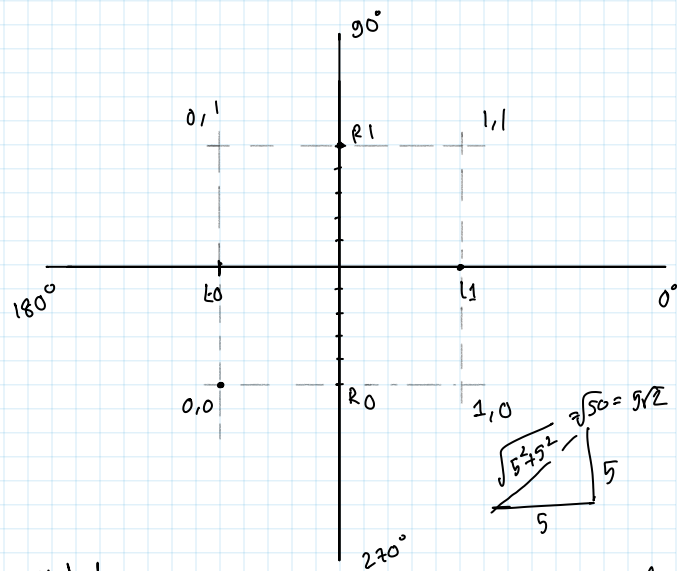
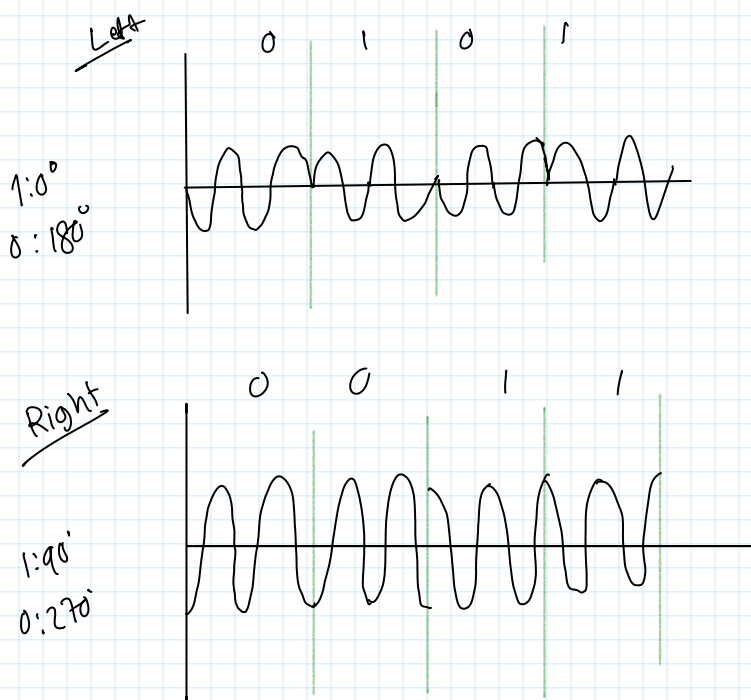
↳ for this : use voltage controlled oscillator

IMPLEMENTATION OF QPSK

00100111

00 10 01 11
 $\overline{L} \overline{R}$ $\overline{L} \overline{R}$ $\overline{L} \overline{R}$ $\overline{L} \overline{R}$

$f: 2$
 $A: 5$



recommended
 range: $-180^\circ \leq \theta \leq 180^\circ$

A

00 $\rightarrow -135^\circ / 225^\circ \rightarrow 5\sqrt{2}$
 01 $\rightarrow 135^\circ \rightarrow 5\sqrt{2}$
 10 $\rightarrow -45^\circ / 315^\circ \rightarrow 5\sqrt{2}$
 11 $\rightarrow 45^\circ \rightarrow 5\sqrt{2}$

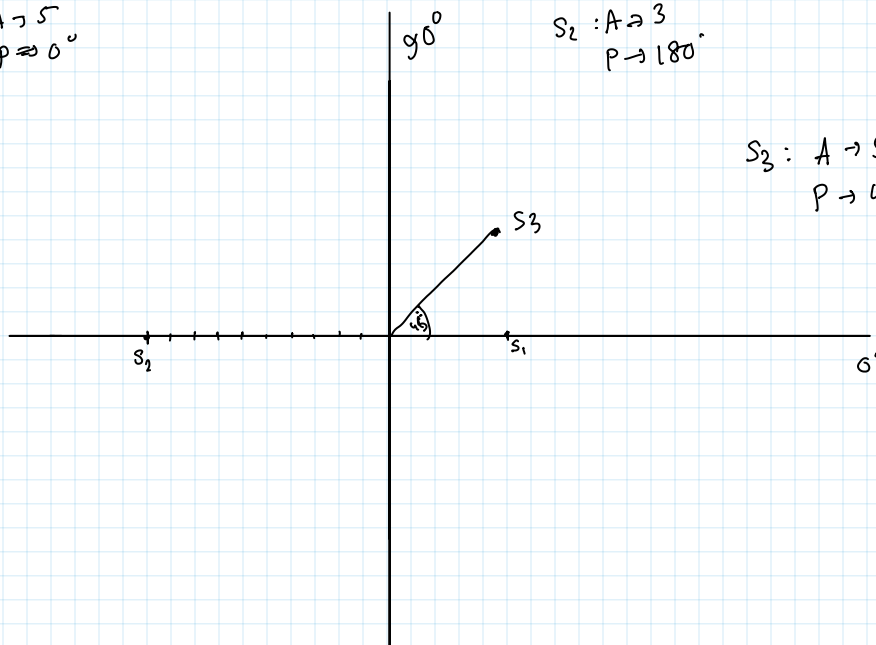
CONSTELLATION DIAGRAM

$r: \text{amp}$
 $\theta: \text{phase}$

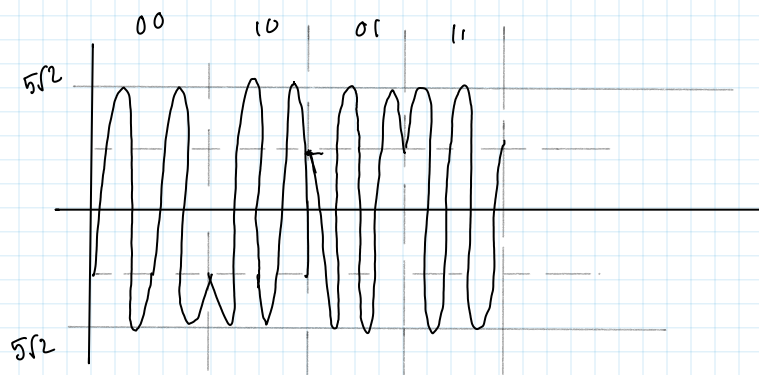
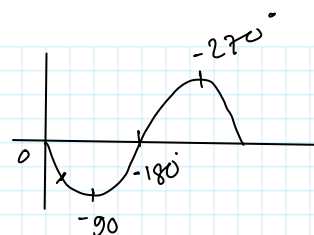
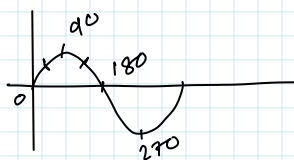
$S_1: A \rightarrow 5$
 $P \rightarrow 0^\circ$

$S_2: A \rightarrow 3$
 $P \rightarrow 180^\circ$

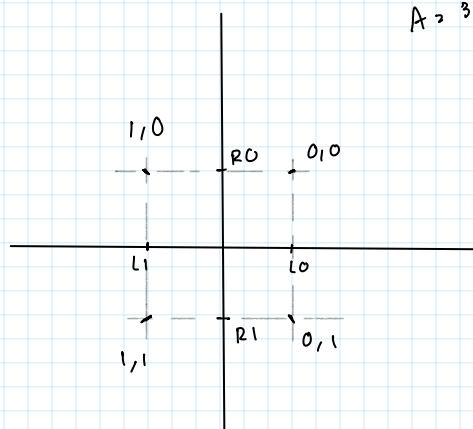
$S_3: A \rightarrow 5$
 $P \rightarrow 45^\circ$



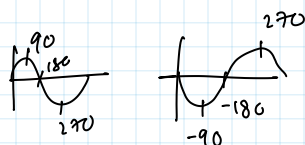
COMBINED SIGNAL



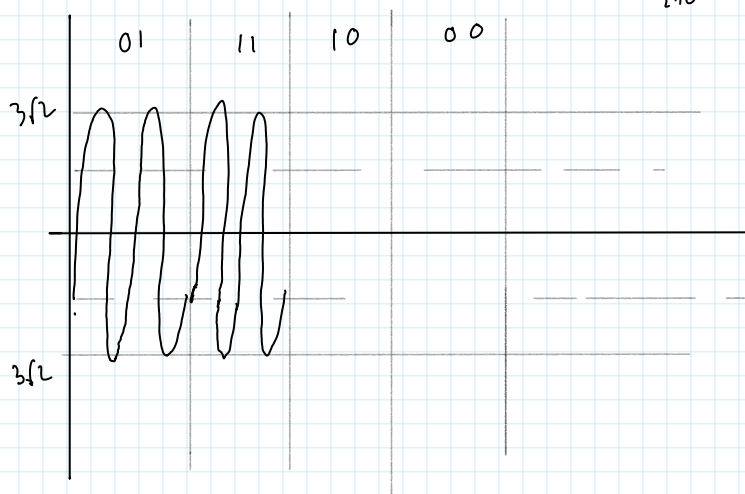
$$A = \sqrt{3^2 + 3^2} = \sqrt{18} = 3\sqrt{2}$$



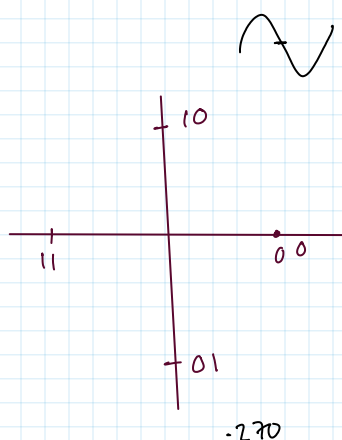
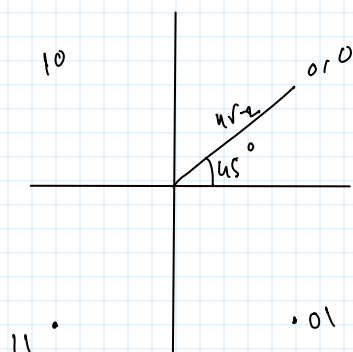
Bits	Phase	A
00	45°	
10	135°	$3\sqrt{2}$
11	-135°	
01	-45°	



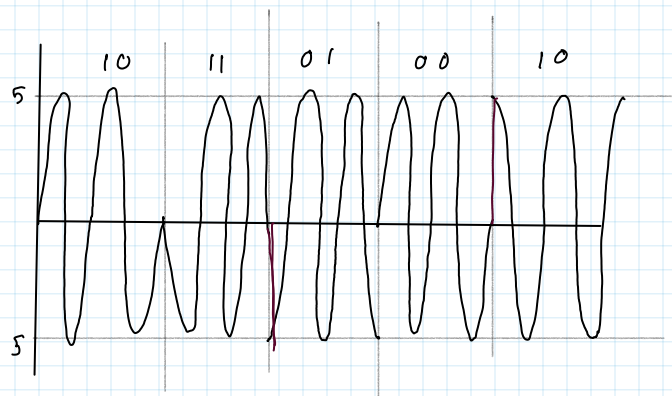
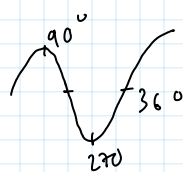
modulated



given cartesian



$00 \rightarrow 0^\circ$



	A	P	F
ASK $\rightarrow$	?	✓	✓
FSK $\rightarrow$	✓	✓	?
PSK $\rightarrow$	✓	?	✓

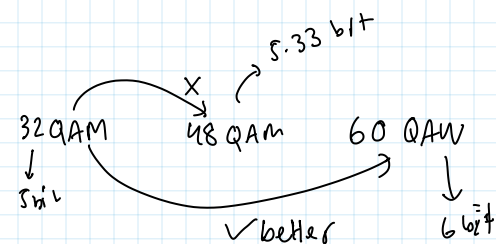
QAM

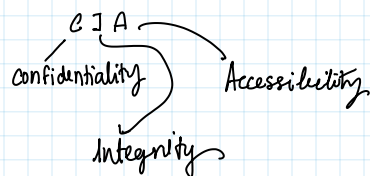
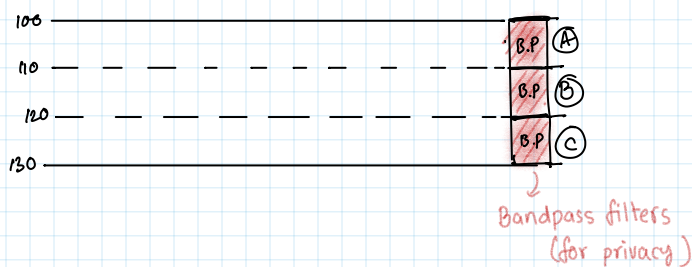
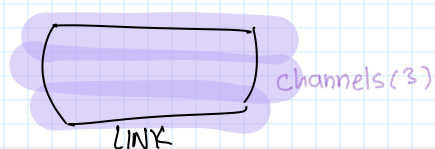
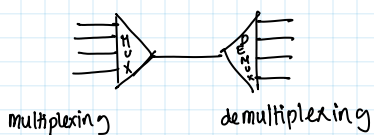
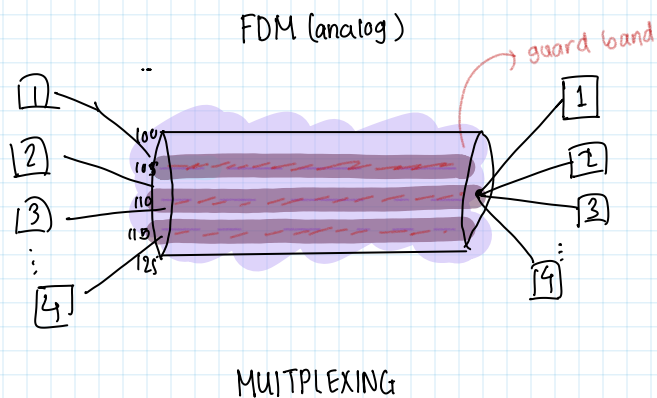
ASK $\rightarrow$ PSK
 $\begin{pmatrix} 2V & 0^\circ \\ 4V & 180^\circ \end{pmatrix}$

$2, 0 \rightarrow 00$
 $4, 0 \rightarrow 01$
 $2, 180 \rightarrow 10$
 $4, 180 \rightarrow 11$

$2 \text{ bit} \rightarrow 4 \text{ QAM}$
 $3 \text{ bit} \rightarrow 8 \text{ QAM}$

QPSK ***
 FSK + PSK + ASK **
 QAM * $\rightarrow$ short





* 5 channels $\rightarrow$ 100 kHz

* g.band $\rightarrow$ 10 kHz

- Total bandwidth required?
- If starting frequency is 600 kHz, ending f?
- show the figure/freq. dom. representation?

(a) $5 \times 100 + 4 \times 10 = 540$

(b) $600 + 540 = 1140$



* total b/w 1180 kHz

each channel 100 kHz

g.band = 10% of each channel

- How many channels?
- Show figure. (assume a value if starting freq. not given)

(a) $100 \times 0.10 = 10$ g.band

$$n \cdot 100 + (n-1) 10 = 1180$$

$$100n + 10n = 1190$$

$$10(10n + n) = 1190$$

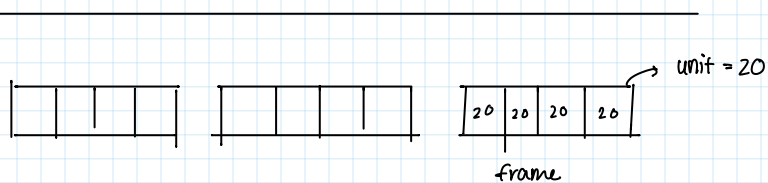
$$110n = 1190$$

$$n = 10$$

ANALOG HIERARCHY

	<u>channel</u>	<u>bandwidth</u>
channel	-	4 kHz
group	12	$12 \times 4 = 48 \text{ kHz}$
supergroup	$5 \times 12 = 60$	$5 \times 48 = 240 \text{ kHz}$
master	$10 \times 60 = 600$	$10 \times 240 = 2400 \text{ kHz}$
jumbo	$6 \times 600 = 3,600$	$6 \times 2.52 = 15.12 \text{ MHz}$

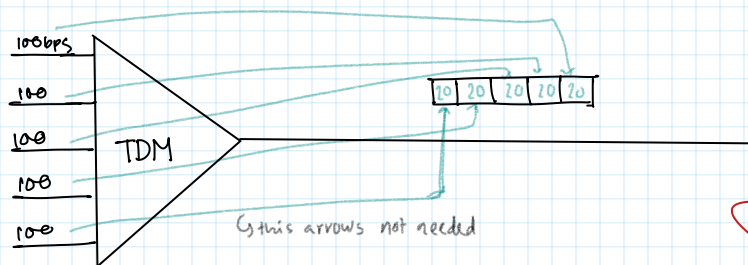
$\left(\frac{2520 - 2400}{9} \right)$ ← $\frac{1}{2}$ band used at input so at supergroup
 $6 \times 2.52 = 15.12$



15/12/25

i/p = 5 $\rightarrow$ i/p rate = 100 bps

* Digital i/p

* unit $\rightarrow$ 20

Frame # of slots = # of input channels

FIGURE MUST
be it mentioned or not

• frame size = # of slots $\times$ unit

• frame rate (কতগুলো ফ্রেম আসছে?)

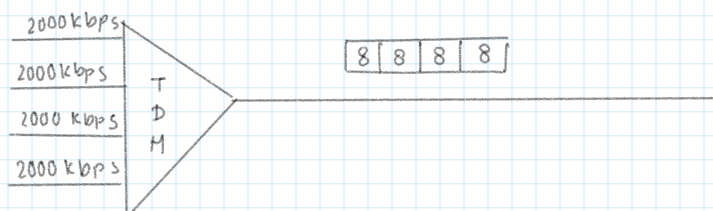
• frame duration = $\frac{1}{\text{frame rate}}$

• output rate = i/p rate $\times$ number of inputs (not usable, explained later)
= frame size $\times$ frame rate

Sample

* 4 i/p channels, each 2 kbps

* multiplexing unit = 1 byte = 8 bits



• Frame size = # of slots $\times$ unit = $4 \times 8 = 32$ bits

• Frame rate = $\frac{2000}{8} = 250$ fps $\left[\frac{\text{i/p rate}}{\text{units}} \right]$

• frame duration = $\frac{1}{\text{frame rate}}$

• o/p rate (Output rate) = frame size $\times$ frame rate
= $(32 \times 250) = 8000$ bps = 8 kbps
Bitrate /
Output Bitrate /
Bitrate in the link

= $\frac{1}{250}$ s

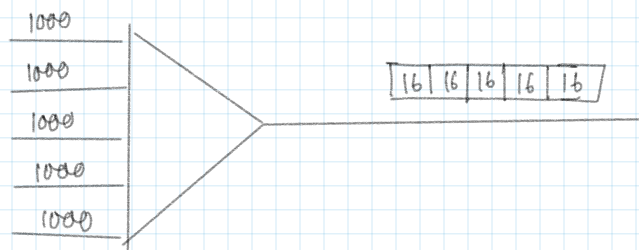
• i/p rate = 2000 bps

• i/p bit duration = $\frac{1}{2000}$ bps = 5×10^{-4} s

• output bit duration = $\frac{1}{8000}$ s

5 i/p channels, 1 kbps

multiplexing unit = 2 bytes = 16 bits



• frame size = $16 \times 5 = 80$ bits

• frame rate = $\frac{1000 \text{ bps}}{16} = 62.5$ fps

• o/p rate = $80 \times 62.5 = 5000$

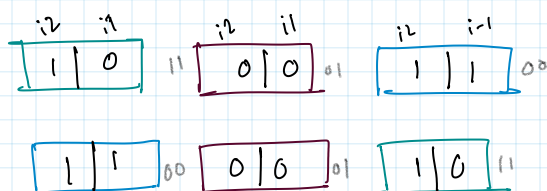
• i/p duration = $\frac{1}{1000} = 1 \times 10^{-3}$

• frame duration = $\frac{1}{62.5}$

• o/p bit duration = $\frac{1}{5000}$

• slot duration = o/p bit duration $\times$ unit

= $\frac{1}{5000} \times 16$



i1: 001

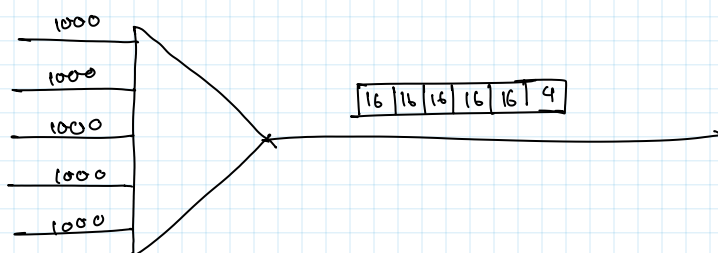
i2: 101

i1: 100
i2: 101

that's why
synchronization bits

— synchronization bits

So now,



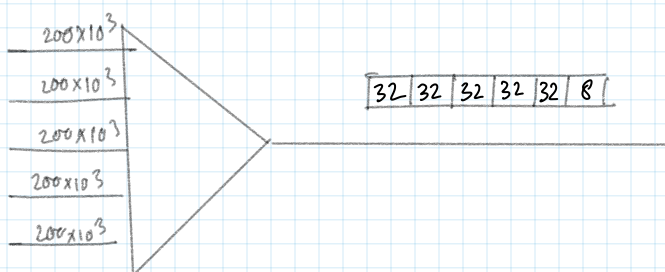
Frame size = $5 \times 16 + 4 = 84$ bits

Frame rate = $\frac{\text{i/p rate}}{\text{unit}} = \frac{1000}{16} = 62.5$ fps

o/p rate = f.size $\times$ f. rate = $84 \times 62.5 = 5250$ bps

* 5 i/p channels $\rightarrow 200 \text{ kbps}$

* multiplexing unit $\rightarrow 4 \text{ bytes} = 4 \times 8 = 32 \text{ bits}$ $\text{sign size} = 1 \text{ byte}$



1 word = 4 bits

$$\text{frame size} = (32 \times 5) + 8 = 168$$

$$\text{fr. rate} = \frac{200 \times 10^3}{32} = 6250$$

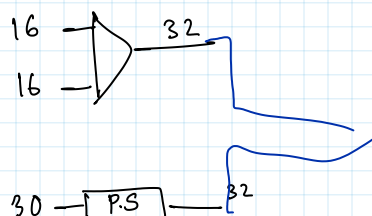
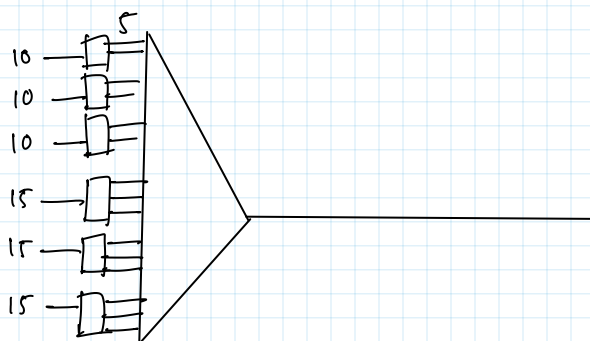
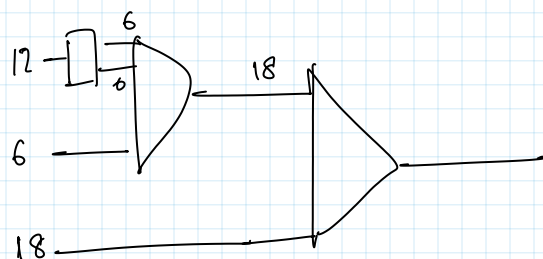
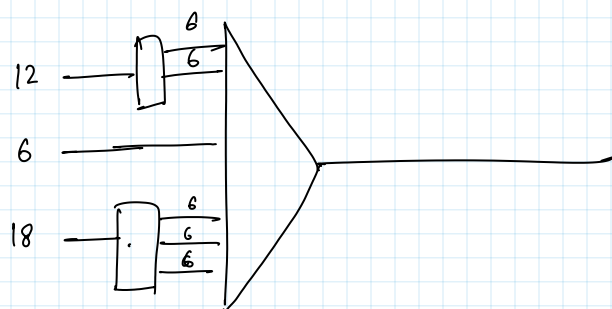
$$\text{o/p rate} = 6250 \times 168 = 1.05 \times 10^6 \text{ bps}$$

$$\bullet \text{ i/p duration} = \frac{1}{200 \times 10^3} = 5 \times 10^{-6} \text{ s}$$

$$\bullet \text{ frame " } = \frac{1}{6250} = 0.00016 \text{ s}$$

$$\bullet \text{ o/p " } = \frac{1}{1.05 \times 10^6} = 9.52 \times 10^{-7} \text{ s}$$

$$\begin{aligned} \bullet \text{ slot duration} &= 9.52 \times 10^{-7} \times 32 \\ &= 3.047 \times 10^{-5} \text{ s} \end{aligned}$$



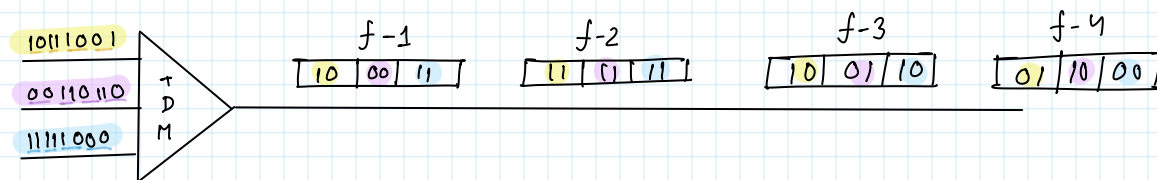
adding dummy bit

Pulse Stuffing

input

unit = 2 ; show the frames.

i-1	i-2	i-3
-----	-----	-----

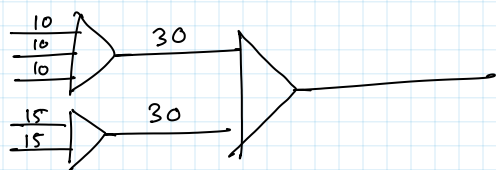
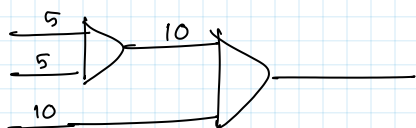


$$f. \text{ size} = 3 \times 2 = 6 \text{ bits}$$

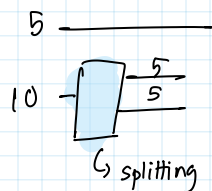
$$f. \text{ rate} = \frac{8}{2} = 4 \text{ fps}$$

$$o/p \text{ rate} = 6 \times 4 = 24 \text{ bps}$$

Multi-level

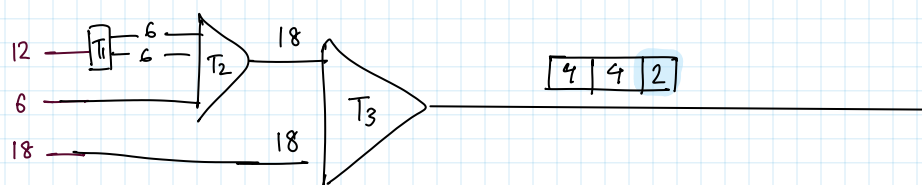


Multiple slot



interleaved bits = 4

syn = 2

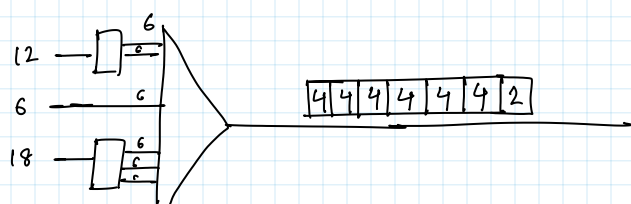


$$f. \text{ size} = 2 \times 4 + 2 = 18$$

$$f. \text{ rate} = \frac{18}{4} = 4.5 \text{ fps}$$

$$o/p \text{ rate} = 18 \times 4.5 = 45 \text{ bps}$$

alternative



$$f. \text{ size} = 6 \times 4 + 2 = 26$$

$$f. \text{ rate} = \frac{6}{4} = 1.5$$

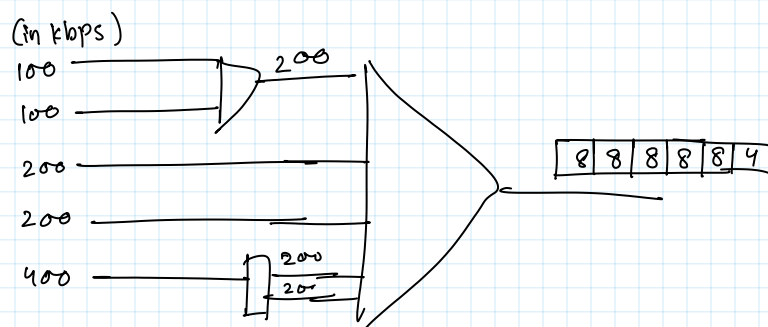
$$o/p = 26 \times 1.5 = 39$$

Multi-level : #no of channel ↓

Multiple-slot : #no. of channels ↑

unit = 1 char = 8 bit

syn = 1/2 char = 4 bit



$$\text{frame size} = (8 \times 8) + 4 = 44$$

$$\text{frame rate} = \frac{200}{8} = 25$$

$$\text{o/p unit} = 44 \times 25 =$$

☐ If unit not mentioned $\rightarrow$ Default unit = 1 bit

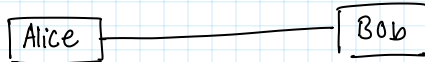
Discard slides : 46 - 49

Slide 60 ✓

80% up $\rightarrow$ synchronous,

30% up $\rightarrow$ statistical.

CHAPTER 10 → Error Detection & Correction



two methods

Detection → detection & retransmission

Correction → detect & correct

error
Wired → 0.001% ⇒ Detection used

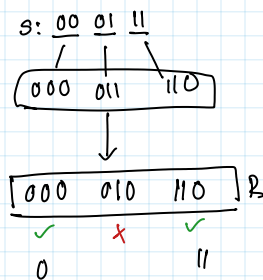
Wireless → more errorprone ⇒ Correction used
~0.1%.

Single bit error → a bit error in a frame

Burst Error → multiple bits error in a frame

Table 10.1 A code for error detection (Example 10.2)

actual data	Datawords	Codewords	what we will send
	00	000	
	01	011	
	10	101	
	11	110	



#table is smartly built such that.

#this table is only for single bit error.

(For Short Question 5)

$$k = \text{len}(dw) = 2$$

$$r(\text{redundant bits}) = 1 \text{ bit(s)}$$

$$n = \text{len}(cw) = 3$$

$$n = k + r$$

Ex: $C(5, 6)$
 $\nearrow$ codeword
 $\searrow$ dataword

$$* \text{len}(dw) = 5$$

$$* \text{len}(cw) = n = 5$$

$$* \text{len}(\text{red}) = r = 2$$

How many codewords are possible in this scheme? Ans: # of codewords = $2^n = 2^5 = 32$

How many of them are valid # valid = # dataword = $2^3 = 8$

Invalid? # invalid = # codewords - # valid
 $= 32 - 8$

$$dw = k = 4$$

$$cw = n = 7$$

$$r = 3$$

$$\# cw = 2^7 =$$

$$\# dw = 2^4 =$$

$\therefore$ valid

$$\therefore \text{invalid} = 2^7 - 2^4 =$$

$$\text{Pairs} = {}^nC_2$$

Hamming distance = distance between 2 codewords

$$(000, 011) = 2$$

$$(000, 101) = 2$$

$$(000, 110) = 2$$

$$(011, 101) = 2$$

$$(011, 110) = 2$$

$$(101, 110) = 2$$

$$H.D = 2$$

Table 10.1 A code for error detection (Example 10.2)

Datawords	Codewords
00	000
01	011
10	101
11	110

Table 10.2 A code for error correction (Example 10.3)

Dataword	Codeword
00	00000
01	01011
10	10101
11	11110

$$(00000, 01011) = 3$$

$$(00000, 10101) = 3$$

$$(00000, 11110) = 4$$

$$(01011, 10101) = 4$$

$$(01011, 11110) = 3$$

$$(10101, 11110) = 3$$

$$\therefore H.D = 3$$

$$\therefore \text{Detection Capability, } d = H.D - 1$$

(guaranteed)
Can do more but sometimes.

$$= 3 - 1 = 2$$

$$\therefore \text{Correction Capability} = \frac{H.D - 1}{2}$$

With the same table

a) What % of CW are invalid? $\text{invalid} = 2^5 - 4 = 28$ $\therefore \text{Percent} = \frac{28}{32} \times 100 = 87.5\%$

b) What is the error detection capability? $H.D - 1 = 2$

c) " " " correction " ? $\frac{3-1}{2} = \frac{2}{2} = 1$

d) you received 1010101010

- any error?

- can you recover? show. 10 01

$$\begin{array}{cc} 10101 & 01010 \\ \underline{10} & \underline{01} \end{array}$$

error (check if within correction capability)
closest: 01011 $\Rightarrow$ 01

e) You received 10001. can you recover?

$$10\bar{0}01 \Rightarrow 10$$

H.D always odd good

$$H.D = 5$$

$$t = \frac{5-1}{2} = 2$$

$$H.D = 6$$

$$t = \frac{6-1}{2} = 2$$

$$H.D = 2 \quad \text{H} \quad t = \frac{2-1}{2} = \frac{1}{2} = 0$$

no guarantee

We use algorithms to make the tables -



CRC - cyclic redundancy code

CRC

* Dataword, $D = 110101$

* Generator, $G = 1001$
(divisor)

Q. why is the generator shared?

S-1:

original dataword + (# of generator bits - 1) empty bits

Augmented Data word = 110101 0 0 0

S-2:

augmented dataword $\div$ divisor

$$\begin{array}{r}
 1001 \overline{) 110101000} \\
 \underline{1001} \\
 01000 \\
 \underline{1001} \\
 0001100 \\
 \underline{1001} \\
 01010 \\
 \underline{1001} \\
 0011
 \end{array}$$

$\therefore$ Remainder = 011

division

dividend $\oplus$ divisor = remainder

S-3: Put remainder in the empty slots in augmented bits.

1101010 1 1

$\rightarrow$ this goes to the receiver

d/w	remainder
-----	-----------

receiver just discards the remainder part to get the data

$C(n, k)$

$C(11, 7)$

Dataword, $D = 101101$

Generator, $G = 1101$

$$\begin{array}{r}
 1101 \overline{) 101101000} \\
 \underline{1101} \\
 01100 \\
 \underline{1101} \\
 0001100 \\
 \underline{1101} \\
 00010
 \end{array}$$

$\therefore \text{remainder} = 010$

$\therefore \text{CodeWord} = 101101010$

Dataword, $D = 101001$

Generator = 1001

$$\begin{array}{r}
 1001 \overline{) 101001000} \leftarrow \text{Augmented word} \\
 \underline{1001} \\
 01101 \\
 \underline{1001} \\
 01000 \\
 \underline{1001} \\
 000100
 \end{array}$$

remainder = 100

$\therefore \text{CodeWord} = 101001100$

MSB

LSB

Error @ 4th bit from LSB

101001100
 4 3 2 1

101000100

MSB $\rightarrow 5$

101001100
1 2 3 4 5

101011100

Receiver

At receiver

codeword
generator

↳ if remainder 0 $\Rightarrow$ all ok
else: error

Ex: 101000100

$$\begin{array}{r}
 1001 \overline{) 101000100} \\
 \underline{1001} \\
 01100 \\
 \underline{1001} \\
 01011 \\
 \underline{1001} \\
 01000 \\
 \underline{1001} \\
 0001
 \end{array}$$

↳ remainder = 1 ; error exists

Ex: 101001100

$$\begin{array}{r}
 1001 \overline{) 101001100} \\
 \underline{1001} \\
 01101 \\
 \underline{1001} \\
 01001 \\
 \underline{1001} \\
 00000
 \end{array}$$

remainder = 0
no error

CR-Polynomial

$$D = 110101$$

$$G = 1001$$

Treat a binary number as coefficients of polynomials

$$101101 \rightarrow x^0$$

$$1 \cdot x^6 + 0 \cdot x^4 + 1 \cdot x^3 + 1 \cdot x^2 + 0 \cdot x + 1 \cdot x^0$$

$$\begin{array}{cccccccc} & & & & & & 7 & \\ 1 & 0 & 0 & 0 & 0 & 1 & 0 & \rightarrow x^7 + x \\ 1 & 1 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 1 & \rightarrow x^{12} + x^{11} + x^7 + x^6 + x^5 + x^3 + x + 1 \\ 12 & 11 & 10 & 9 & 8 & 7 & 6 & 5 & 4 & 3 & 2 & 1 \end{array}$$

$$x^{11} + x^5 + x \rightarrow 100000100010$$

Polynomial Division

$$D = 110101 \rightarrow x^5 + x^4 + x^2 + 1$$

$$G = 1001 \rightarrow x^3 + 1$$

S-1: Multiply the dividend with the max degree of division

$$\begin{aligned} \text{Augmented D/N} &= x^3(x^5 + x^4 + x^2 + 1) \\ &= x^8 + x^7 + x^5 + x^3 \end{aligned}$$

S-2: polynomial division.

$$\begin{array}{r} x^3+1 \overline{) \begin{array}{l} x^8+x^7+x^5+x^3 \\ x^8+x^5 \end{array} } \quad x^5+x^4+x+1 \\ \hline x^3+1 \overline{) \begin{array}{l} x^7+x^3 \\ x^7+x^4 \end{array} } \\ \hline x^3+1 \overline{) \begin{array}{l} x^4+x^3 \\ x^4+1 \end{array} } \\ \hline x^3+1 \overline{) \begin{array}{l} x^3+1 \\ x^3+1 \end{array} } \\ \hline 0 \end{array}$$

S-3: Add the augmented word
(zero in this case)

$$D = 101101 \rightarrow x^5 + x^3 + x^2 + 1$$

$$G = 1101 \rightarrow x^3 + x^2 + 1$$

S-1:

$$\begin{aligned} \text{Aug. D/W} &= x^3(x^5 + x^3 + x^2 + 1) \\ &= x^8 + x^6 + x^5 + x^3 \end{aligned}$$

S-2:

$$\begin{array}{r} x^3+x^2+1 \overline{) \begin{array}{l} \cancel{x^8} + \cancel{x^6} + \cancel{x^5} + x^3 \\ \cancel{x^8} + \cancel{x^7} + \cancel{x^6} \end{array}} \quad x^5 + x^4 + x \\ \hline x^3+x^2+1 \overline{) \begin{array}{l} \cancel{x^7} + \cancel{x^6} + x^3 \\ \cancel{x^7} + \cancel{x^6} + x^4 \end{array}} \\ \hline x^3+x^2+1 \overline{) \begin{array}{l} \cancel{x^4} + \cancel{x^3} \\ \cancel{x^4} + \cancel{x^3} + x \end{array}} \\ \hline x \end{array} \quad \text{Remainder} = x$$

$$\therefore C/W = x^8 + x^6 + x^5 + x^3 + x$$

Error

$$\begin{array}{l} x^8 + x^6 + x^5 + x^3 + x \\ \swarrow \text{LSB 4th} \rightarrow x^8 + x^6 + x^5 + x \\ \searrow \text{LSB 5th (error)} \rightarrow x^8 + x^6 + x^5 + x^4 + x^3 + x \end{array}$$

→ we'll get remainder on the receiver side when error

$$\begin{array}{r} \underline{x^3+x^2+1} \overline{) \begin{array}{l} \cancel{x^8} + \cancel{x^6} + \cancel{x^5} + x \\ \cancel{x^8} + \cancel{x^7} + \cancel{x^6} \end{array}} \quad x^5 + x^4 + x \\ \hline x^3+x^2+1 \overline{) \begin{array}{l} \cancel{x^7} + \cancel{x^6} + x \\ \cancel{x^7} + \cancel{x^6} + x^4 \end{array}} \\ \hline x^3+x^2+1 \overline{) \begin{array}{l} \cancel{x^4} + x \\ \cancel{x^4} + \cancel{x^3} + x \end{array}} \\ \hline x^3 \\ \hline x^3+x^2+1 \\ \hline \textcircled{2} \rightarrow \text{error detection} \end{array}$$

$$\begin{array}{r|l}
 x^3+x^2+1 & x^8+x^6+x^5+x^4+x^3+x \\
 \hline
 & x^8+x^7+x^6 \\
 \hline
 & x^7+x^5+x^4+x^3+x \\
 & x^3+x^2+1 \\
 \hline
 & x^7+x^6+x^4 \\
 & x^3+x^2+1 \\
 \hline
 & x^3+x \\
 & x^3+x+1 \\
 \hline
 & 1
 \end{array}
 \quad \left| \quad x^5+x^4+1 \right.$$

Receiver $\rightarrow$

$$G = x^3+1$$

1. check for error (check for remainder zero)
2. remove the terms from C/W that are lower than the highest degree of G & then divide them with that (i.e. x^3) to reverse the augmentation part

CRC

$\rightarrow$ costly

$\rightarrow$ when security not much imp: check sum

S-1:	0	11	7	3	5	<u>sum</u>
						35
R-1:	9	11	6	3	5	35
						34x

this is not much strong

Q. why check sum not strong

S:	9	11	7	3	5	} same sum
R:	9	10	8	3	5	

$$\begin{array}{r}
 10 \rightarrow 1010 \\
 -10 \rightarrow 0101 \\
 \hline
 1111 \\
 0000 \text{ is complement}
 \end{array}$$

S:	9	11	7	3	-30
	1001	1011	0111	0011	

$$\begin{array}{r}
 27: 111011 \\
 27: 1011 \\
 \hline
 1100 \\
 -27: 0011
 \end{array}$$

28.